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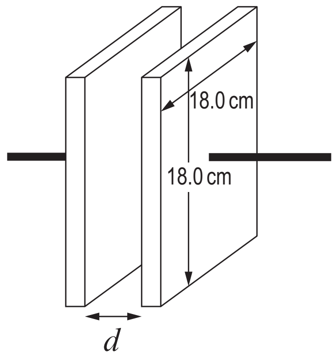
SECTION A

Answer **all** *questions.*

1. *(a)* (i) A laboratory technician has two square aluminium plates with sides of length 18.0 cm. Calculate the separation, d, of the plates that produces a capacitance of 2.0 nF. [3]



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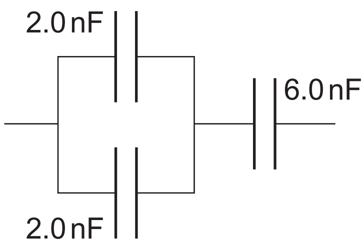
(ii) Calculate the capacitance of the capacitor combination shown. [3]



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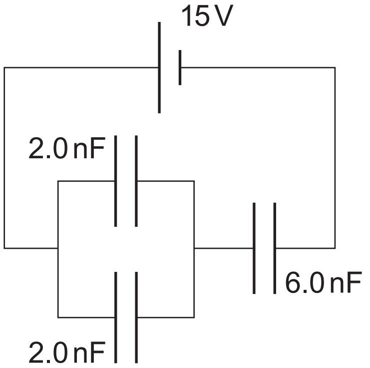
(iii) Show that the charge stored on a 2.0 nF capacitor in the following circuit is 18 nC. [3]



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(iv) Calculate the total energy stored by the capacitor combination. [2]

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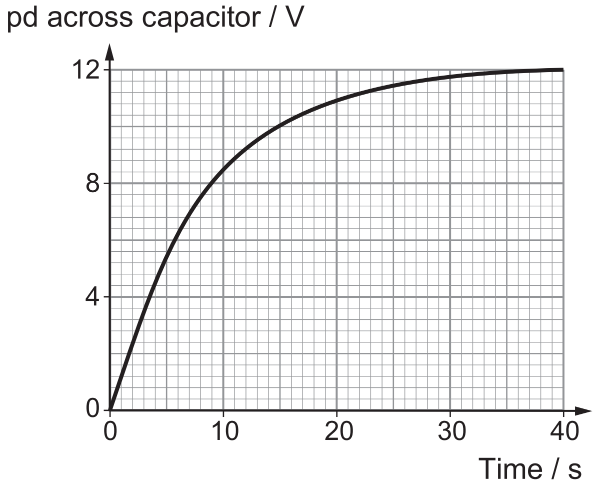
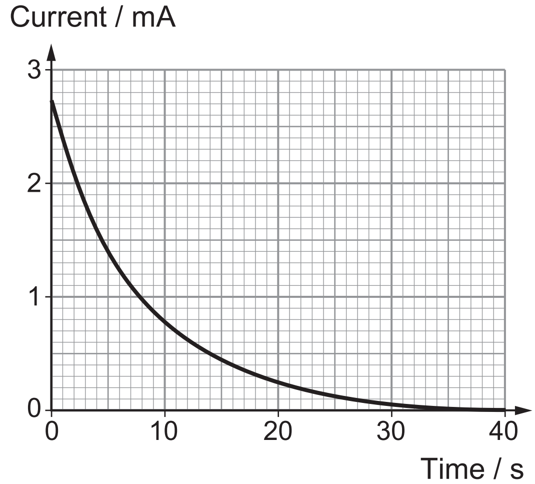
(b) A group of students investigated the **charging** of a capacitor through a resistor and obtained the following data.





(i) Draw a circuit diagram of the apparatus that might have been used to obtain the data. [2]

(ii) Use the graphs to determine:

- the emf of the power supply;
- the resistance of the resistor;
- the capacitance of the capacitor. [5]

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